

2.4 Exercise Sheet 4

2.4.A Numerical and Exploration

Exercise 2.4.1. Here we list some pairs (E, P) , where $E/\mathbb{Q}$ is an elliptic curve and $P \in E(\mathbb{Q})$. Using Sage, find the order of P in $E(\mathbb{Q})$.

- (a) $E : y^2 = x^3 - 4x + 4$ and $P = (2, 2)$.
- (b) $E : y^2 = x^3 - x + 1$ and $P = (1, 1)$.
- (c) $E : y^2 = x^3 - 219x + 1654$ and $P = (11, 24)$.
- (d) $E : y^2 + xy = x^3 - x + 137$ and $P = (2, 11)$.
- (e) $E : y^2 = x^3 - 189x + 1269$ and $P = (3, 27)$. Also do $P = (21, 81)$ on the same curve.

Find other interesting examples on the LMFDB.

Exercise 2.4.2. (Adapted from [2] Exercise 2.12]) Using the Nagell-Lutz Theorem or otherwise, for each elliptic curve $E/\mathbb{Q}$ below, find all rational points of finite order on E and determine the structure of $\text{Tors } E(\mathbb{Q})$:

- (a) $y^2 = x^3 - 2$
- (b) $y^2 + xy + y = x^3$.
- (c) $y^2 = x^3 + 1$
- (d) $y^2 = x^3 - 43x + 166$
- (e) $y^2 + xy = x^3 - 45x + 81$

(Hint: You may have to complete the square to bring the given curve into shortest Weierstrass form to apply Nagell-Lutz. At least try to do some of these computations by hand (i.e., without using Sage) to convince yourself that they can be done in a reasonable amount of time!)

Exercise 2.4.3. Let p be a prime and let $E/\mathbb{Q}$ be the elliptic curve with equation $y^2 = x^3 + p^2$. What is $\text{Tors } E(\mathbb{Q})$? Experiment in Sage, make conjectures, and then prove your conjectures.

Exercise 2.4.4.

- (a) Fix a prime p . For each integer $m \in \mathbb{Z}_{\geq 0}$, let $S_m(p) := \sum_{x=0}^{p-1} x^m$. What can you say about the function $m \mapsto S_m(p) \pmod{p}$? Observe patterns, and make and prove conjectures. What does the value of 0^0 need to be in order to obtain a very clean and uniform result?
- (b) (Only for those who know about finite fields other than $\mathbb{F}_p$ for primes p .) Fix a finite field $\mathbb{F}_q$. Repeat the same as in (a) for the function $m \mapsto \sum_{x \in \mathbb{F}_q} x^m$.

Exercise 2.4.5 (Fisher). Let p be an odd prime, and let E be an elliptic curve over $\mathbb{F}_p$.

- (a) Show that there exists an elliptic curve E' over $\mathbb{F}_p$ such that $\#E(\mathbb{F}_p) + \#E'(\mathbb{F}_p) = 2(p+1)$. (Hint: Modulo the right definitions of these objects, you could probably have solved this problem before you knew anything about elliptic curves.)
- (b) Given the curves E and E' as in (a), what do you notice about the sizes of $E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$ and of $E(\mathbb{F}_{p^2})$ and $E'(\mathbb{F}_{p^2})$? Observe patterns and make conjectures.
- (c) (**) Prove your conjectures from (b). (Hint: It might be helpful to consider the “conjugation” map $\mathbb{F}_{p^2} \rightarrow \mathbb{F}_{p^2}$ given by $a \mapsto a^2$. If you don’t know much about the fields $\mathbb{F}_{p^2}$ in general, try to attempt this problem for small special cases such as $p = 3$ where $\mathbb{F}_9 \cong \mathbb{F}_3[i] := \mathbb{F}_3[x]/(x^2 + 1)$ or $p = 5$ where $\mathbb{F}_{25} \cong \mathbb{F}_5[\omega] := \mathbb{F}_5[x]/(x^2 + x + 1)$.)

Exercise 2.4.6. Let K be a field and $E \subset \mathbb{P}_K^2$ be an elliptic curve. In §1.12.E we studied what the group $E[3](\overline{K})$ can look like, where $\overline{K}$ denotes an algebraic closure of K . Perform an analysis similar to the one in §1.12.C to figure out what happens when K is not necessarily algebraically closed. For instance:

- (a) Give an example of a field K and an elliptic curve E/K such that $(\#E[3](\overline{K}), \#E[3](K))$ is (i) $(9, 1)$ (ii) $(9, 3)$, (iii) $(9, 9)$, (iv) $(3, 1)$, (v) $(3, 3)$, and (vi) $(1, 1)$.
- (b) (**) As was done in §1.12.C classify exactly when each of these possibilities in (a) occur.
- (c) (*) Show that for any field K , there is an extension L/K of degree $[L : K] \leq 8$ such that there is a $P \in E[3](L) \setminus \{\mathcal{O}\}$. Show that if $\text{ch } K = 3$ and E is not supersingular, this bound can be improved to $[L : K] \leq 3$. Give examples to show that both of these bounds are sharp. (Hint: Look at the proof of 1.12.22 carefully, and use that $8 = 4 \cdot 2$.)

2.4.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.4.7. Let K be a field, and for $\lambda \in K \setminus \{0, 1\}$, let

$$j(\lambda) := 2^8 \frac{(\lambda(1-\lambda)+1)^3}{\lambda^2(1-\lambda)^2}.$$

- (a) For $\lambda, \lambda' \in K \setminus \{0, 1\}$, we have that $j(\lambda) = j(\lambda')$ iff $\lambda' \in S(\lambda)$, where

$$S(\lambda) := \left\{ \lambda, \frac{1}{\lambda}, 1-\lambda, \frac{1}{1-\lambda}, 1-\frac{1}{\lambda}, 1-\frac{1}{1-\lambda} \right\}.$$

- (b) For any $\lambda \in K \setminus \{0, 1\}$, the set $S(\lambda)$ of (a) has cardinality exactly six.

Exercise 2.4.8. Let K be a field.

- (a) For any elliptic curve E over K , there is a(n) (admissible) change of coordinates which brings E into the form $E_\lambda : y^2 = x(x-1)(x-\lambda)$ for some $\lambda \in K \setminus \{0, 1\}$. The j -invariant of E_λ is precisely the $j(\lambda)$ of 2.4.7 i.e., $j(E_\lambda) = j(\lambda)$.
- (b) For $\lambda, \lambda' \in K \setminus \{0, 1\}$, there is an admissible change of coordinates taking E_λ to $E_{\lambda'}$ iff $\lambda' \in S(\lambda)$ iff $j(\lambda) = j(\lambda')$. In particular, the j -invariant classifies elliptic curves up to admissible changes of coordinates (and hence by 1.10.6 up to isomorphism). (Hint: The “change in the x -coordinate alone” would have to send the unordered triple $\{0, 1, \lambda\}$ to $\{0, 1, \lambda'\}$.)

Exercise 2.4.9. Let K be a field (of any characteristic!). Given any $j \in K$, there is a unique elliptic curve E over K (up to admissible changes of coordinates over K) such that $j(E) = j$.

Exercise 2.4.10. Let K be a field. Given finitely many points $P_i \in \mathbb{P}^2(K)$, there is a line $L \subset \mathbb{P}_K^2$ that doesn't pass through any of the P_i . In particular, up to a linear change of coordinates, we can assume that each P_i lies in $\mathbb{P}_K^2 \setminus \mathbb{V}(Z) \cong \mathbb{A}_K^2$.

Exercise 2.4.11. Let K be a field and $E \subset \mathbb{P}_K^2$ an elliptic curve in Weierstrass form.

- (a) If $P, Q \in E(K)$ are inflection points, then so is $P * Q$.
- (b) If $S \subset E(K)$ is the set of inflection points, then no four points in S are collinear, but any line passing through two points in S passes through a third point in S .

Exercise 2.4.12.

- (a) (*) Suppose $S \subset \mathbb{A}^2(\mathbb{R})$ is a finite set of points with the property that a line passing through any two of them passes through a third. Then S is contained in a line. (Hint: This is just plane geometry; it has nothing to do with elliptic curves.)
- (b) If $E \subset \mathbb{P}_{\mathbb{R}}^2$ is an elliptic curve, then exactly three of its nine complex inflection points are real, i.e., even though $\#E[3](\mathbb{C}) = 9$, the subgroup $E[3](\mathbb{R}) \subset E[3](\mathbb{C})$ has size $\#E[3](\mathbb{R}) = 3$. In particular, $E[3](\mathbb{R}) \cong_{\text{Ab}} \mathbb{Z}/3$. (See also 2.2.9)

Exercise 2.4.13 (Fisher). Let $K = \mathbb{Q}$ and let E be the elliptic curve with Weierstrass equation $y^2 = x^3 + ax^2 + bx$ with $a, b \in \mathbb{Z}$. If $P = (x, y) \in \text{Tors } E(\mathbb{Q}) \setminus \{0\}$, then either $x = 0$ or $x \mid b$ and $x + a + (b/x)$ is a perfect square. (Hint: Review the proof of the Nagell-Lutz Theorem.)

Exercise 2.4.14. ()** Let $E/\mathbb{Q}$ be an elliptic curve given in long Weierstrass form with all coefficients $a_1, \dots, a_6 \in \mathbb{Z}$. If $P = (x, y) \in \text{Tors } E(\mathbb{Q}) \setminus \{\mathcal{O}\}$, then $x, y \in \mathbb{Z}$. (Hint: Consider $E : y^2 + xy = x^3 + 4x + 1$. For a strong salvage, show using techniques very similar to our proof of Nagell-Lutz that $4x, 8y \in \mathbb{Z}$. If you're motivated, you can even try to show that if $2 \mid a_1$ or if $2P \neq \mathcal{O}$, then $x, y \in \mathbb{Z}$.)

For the next few exercises, we use the following notation: if p is a prime, and $E/\mathbb{F}_p$ an elliptic curve, we let $a_p(E) := p + 1 - \#E(\mathbb{F}_p) \in \mathbb{Z}$. The integer $a_p(E)$ is called the *trace of Frobenius* associated to the elliptic curve $E/\mathbb{F}_p$. Note that $\#E(\mathbb{F}_p) = p + 1$ iff $a_p(E) = 0$.

Exercise 2.4.15. Let p be a prime other than 2, and let $E : y^2 = f(x)$ be an elliptic curve over $K = \mathbb{F}_p$ in short Weierstrass form.

- (a) The integer $a_p(E)$ is congruent mod p to the coefficient of x^{p-1} in $f(x)^{(p-1)/2}$. (Hint: The Legendre symbol and 2.4.4)

- (b) Suppose that $f(x) = x(x-1)(x-\lambda)$ for $\lambda \in \mathbb{F}_p \setminus \{0, 1\}$. The coefficient of x^{p-1} in $f(x)^{(p-1)/2}$ is $h_p(\lambda)$, where $h_p(x) \in \mathbb{F}_p[x]$ is the polynomial defined by

$$h_p(x) := \sum_{i=0}^{(p-1)/2} \binom{(p-1)/2}{i}^2 x^i \in \mathbb{F}_p[x].$$

We have that $h_p(0) \cdot h_p(1) \neq 0$.

Exercise 2.4.16. Fix an integer $b \in \mathbb{Z}$.

- (a) For each prime p with $p \nmid 2b$, the equation $y^2 = x^3 + bx$ defines an elliptic curve E_b over $\mathbb{F}_p$. Further,

$$a_p(E_b) \equiv \binom{(p-1)/2}{(p-1)/4} b^{(p-1)/4} \pmod{p}.$$

- (b) (**) Further, if $p \geq 5$ and $p \equiv 3 \pmod{4}$, then $a_p(E_b) = 0$.

Exercise 2.4.17. Fix an integer $c \in \mathbb{Z}$.

- (a) For each prime p , the equation $y^2 = x^3 + c$ defines an elliptic curve E_c over $\mathbb{F}_p$. Further,

$$a_p(E_c) \equiv \binom{(p-1)/2}{(p-1)/3} c^{(p-1)/6} \pmod{p}.$$

- (b) (**) Further, if $p \geq 5$ and $p \equiv 2 \pmod{3}$, then $a_p(E_c) = 0$.

How do 2.4.17 and 2.4.16 relate to 2.1.8? A possible hint for both 2.4.17(a) and 2.4.16(a): experiment in Sage! A possible hint for 2.4.17(b) and 2.4.16(b): combine your solution to (a) with [8, Theorem V.1.1]. Is there an easier way?

Here are a few more exercises, just for fun.

Exercise 2.4.18. Let K be a field. For $h(x) \in K[x]$ and $n \in \mathbb{Z}_{\geq 0}$, let $h^{(n)}(x)$ denote the n^{th} (formal) derivative of h with respect to x , i.e.,

$$h^{(n)}(x) = \frac{d^n}{dx^n} h(x) \in K[x].$$

Let $\bar{K}$ be an algebraic closure of K , and suppose there is an $\alpha \in \bar{K}$ such that $h^{(n)}(\alpha) = 0$ for all $n \geq 0$. Then h is the zero polynomial, i.e., $h(x) = 0 \in K[x]$. In other words, if $h(x) \in K[x]$ is not the zero polynomial, then for each $\alpha \in \bar{K}$, there is some $n \in \mathbb{Z}_{\geq 1}$ such that $h^{(n)}(\alpha) \neq 0$. (Hint: For a strong salvage, consider the case in which either $\text{ch } K = 0$ or $\text{ch } K = p > \deg h$. It may help to “translate” so that $\alpha = 0$.)

Exercise 2.4.19. Let K be a field. A *linear differential operator* $\mathcal{D}$ of degree $n \in \mathbb{Z}_{\geq 1}$ is an expression of the form

$$\mathcal{D} = c_0(x) \frac{d^n}{dx^n} + c_1(x) \frac{d^{n-1}}{dx^{n-1}} + \cdots + c_{n-1}(x) \frac{d}{dx} + c_n(x),$$

where $c_0(x), c_1(x), \dots, c_n(x) \in K[x]$ and $c_0(x) \neq 0$.

- (a) Given a differential operator $\mathcal{D}$ and a polynomial $h(x) \in K[x]$, describe what it means to “apply $\mathcal{D}$ to h ” to get the polynomial $(\mathcal{D}h)(x) \in K[x]$. The resulting map $\mathcal{D} : K[x] \rightarrow K[x]$ is K -linear.
- (b) Suppose $\mathcal{D}$ as above is a differential operator of order $n \in \mathbb{Z}_{\geq 1}$ and suppose $h(x) \in K[x]$ is a nonzero polynomial killed by $\mathcal{D}$, i.e., such that $(\mathcal{D}h)(x) = 0 \in K[x]$. If $\bar{K}$ is an algebraic closure of K and $\alpha \in \bar{K}$ a root of $h(x)$ of multiplicity at least n , then $c_0(\alpha) = 0$. (Hint: 2.4.18.)

Exercise 2.4.20. Let p be a prime other than 2, and let $h_p(x) \in \mathbb{F}_p[x]$ be the polynomial defined in 2.4.15.

- (a) Let $\mathcal{D}$ be the *Picard-Fuchs differential operator*

$$\mathcal{D} := 4x(1-x) \frac{d^2}{dx^2} + 4(1-2x) \frac{d}{dx} - 1.$$

We have that $(\mathcal{D}h_p)(x) = 0 \in \mathbb{F}_p[x]$.

- (b) The polynomial $h_p(x) \in \mathbb{F}_p[x]$ is separable, i.e., has no repeated roots in $\overline{\mathbb{F}}_p$. In particular, there are exactly $(p-1)/2$ elements $\lambda \in \overline{\mathbb{F}}_p \setminus \{0, 1\}$ such that $h_p(\lambda) = 0$.

The following result might explain why this problem is worth considering and potentially motivate you to attempt it: the curve $E : y^2 = x(x-1)(x-\lambda)$ over $\overline{\mathbb{F}}_p$ is supersingular iff $h_p(\lambda) = 0$ ([S Theorem V.4.1(b)]). This in turn allows us to count the number of supersingular elliptic curves over $\overline{\mathbb{F}}_p$ up to isomorphism.